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| Score: / 10 |
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Name (Print): \_\_\_\_\_

Calculus I (MATH 2200)

Spring 2020

Name (Print): \_\_\_\_\_

Wednesday, 02/12/2020

Quiz 3: Section 2.5, 2.6

Name (Print): \_\_\_\_\_

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## Section 2.5

1. Determine the *end behavior* (that is, the limit as  $x \rightarrow \pm\infty$ ) of the following functions:

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| 1 pts |
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(a)  $f(x) = \frac{3x + 2}{x^2 - 1}$

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|-------|
| 1 pts |
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(b)  $g(x) = \frac{40x^4 + 4x^2 - 1}{10x^4 + 8x^2 + 1}$

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| 2 pts |
|-------|

(c)  $g(x) = \frac{91x^7 - x^2 + 2}{1000x^3 + 800x^2}$

2 pts

2. Determine the end behavior of  $f(x) = \frac{10x^2 + x + 1}{x^{2.2} + \sqrt{x}}$ .

Section 2.6

- 2 pts 1. Let  $f(x) = \begin{cases} \frac{x^2 + x - 6}{x - 2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$ . Is  $f(x)$  continuous at  $x = 2$ ? Explain.

2 pts

2. Evaluate the limit  $\lim_{x \rightarrow 2} \cos\left(\frac{x^2 - 4}{x - 2}\right)$ .